higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

16 April 2020

This marking guideline consists of 6 pages.

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QUESTION 1

1.1	1.1.1	D		
	1.1.2	C		
	1.1.3	D		
	1.1.4	B		
	1.1.5	C		
			(5 × 1)	(5)
1.2	1.2.1	positive		
	1.2.2	carry out		
	1.2.3	simultaneously		
	1.2.4	toggle		
	1.2.5	5		
			(5 × 1)	(5)
1.3	1.3.1	E		
	1.3.2	C		
	1.3.3	A		
	1.3.4	B		
	1.3.5	D		
			(5 × 1)	(5)
1.4	1.4.1	True		
	1.4.2	True		
	1.4.3	False		
	1.4.4	False		
	1.4.5	True		
			(5 × 1)	(5)
1.5	1.5.1	bit		
	1.5.2	multi-vibrator		
	1.5.3	latch		
	1.5.4	fan-out		
	1.5.5	nonweighted		
			(5 × 1)	(5)
1.6	1.6.1	full		
	1.6.2	decoder		
	1.6.3	minimum		
	1.6.4	NOR		
	1.6.5	excess-3		
			(5 × 1)	(5)
				[30]

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QUESTION 2

2.1 2.1.1 $-58,75_{10}$ = 0111010, 11.✓
1000101,00

$-46,5_{10}$ = 0101110, 10.✓
1010001,01

1000101,00
+1010001,01
0010110,01.✓
+1
0010110,10.✓
-1101001, 01.✓
= $-105,25_{10}$.✓ (6)

2.1.2 C, A_{16} = 1100, 1010.✓
0011,0101

7, B_{16} = 0111, 1011.✓
1000,0100

0011,0101.✓
+ 1000,0100
1011,1001.✓
-0100, 0110.✓
= $-4,375_{10}$.✓ (6)

2.2 2.2.1 $6A,5_{16}$ = 01101010,0101✓
$3,7_8$ = 011,111✓

11011,011 ✓ = $27,375_{10}$✓
111110 | 11010100101000
-111110
1011000
-111110✓
1101010
-111110
1011001
-111110
1101100✓
-111110
1011100 (6)

2.2.2 $6D, A_{16} = 01101101,1010$✓
$55,5_8 = 101101,101$✓

```
                    11011011010
                      101101101
                  ---------------
                    11011011010✓
                   00000000000
                  11011011010
                 11011011010
                00000000000
               11011011010
              11011011010
             00000000000
            11011011010✓
          ---------------
          1001110001001,1010010✓
```

$= 5001,640625_{10}$✓ (6)

2.3 2.3.1 110110_2

2.3.2
```
  2    8    4
 +3   +3   +3
 ---  ---  ---
  5   11    7
```
5B,7✓
1011011,0111✓

2.3.3 8421 8421 8421 8421
0010 1010 1111 1000✓
101010,1111000✓

(3 × 2) (6)
[30]

QUESTION 3

3.1 3.1.1 $XX + XY(\overline{X} + \overline{Y} + Z)$✓

$X + XY(\overline{X} + \overline{Y} + Z)$✓

$X(1 + Y)(\overline{X} + \overline{Y} + Z)$✓

$X(\overline{X} + \overline{Y} + Z)$✓

$X\overline{X} + X\overline{Y} + XZ$✓

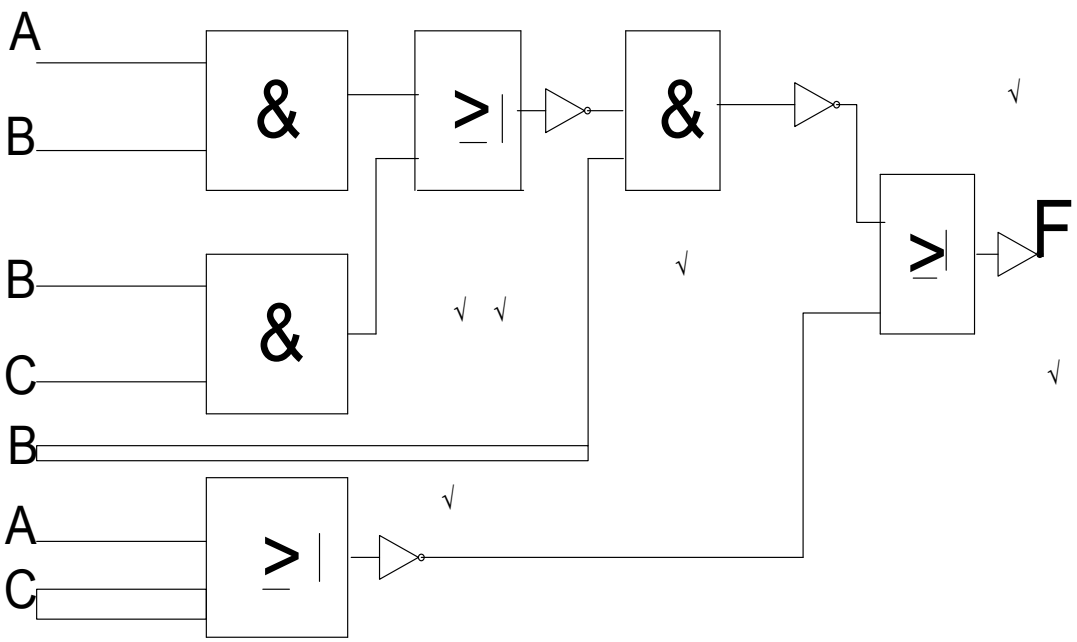
$X\overline{Y} + XZ$✓ (6)

3.1.2 $(\overline{A} + \overline{\overline{A}} + \overline{B})(\overline{B} + \overline{\overline{A}} + \overline{B})$✓

$(\overline{A} + A + B)(\overline{B} + A + B)$✓

$(1 + B)(1 + A)$✓
$=1$✓ (4)

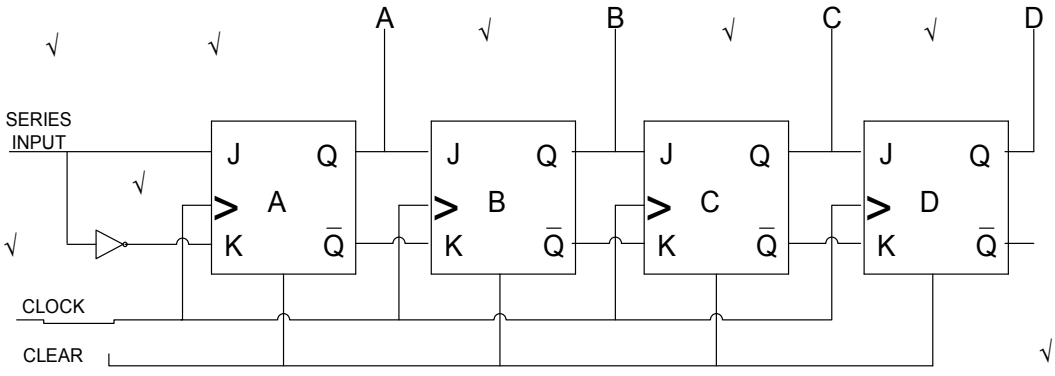
3.2



(6)
[16]

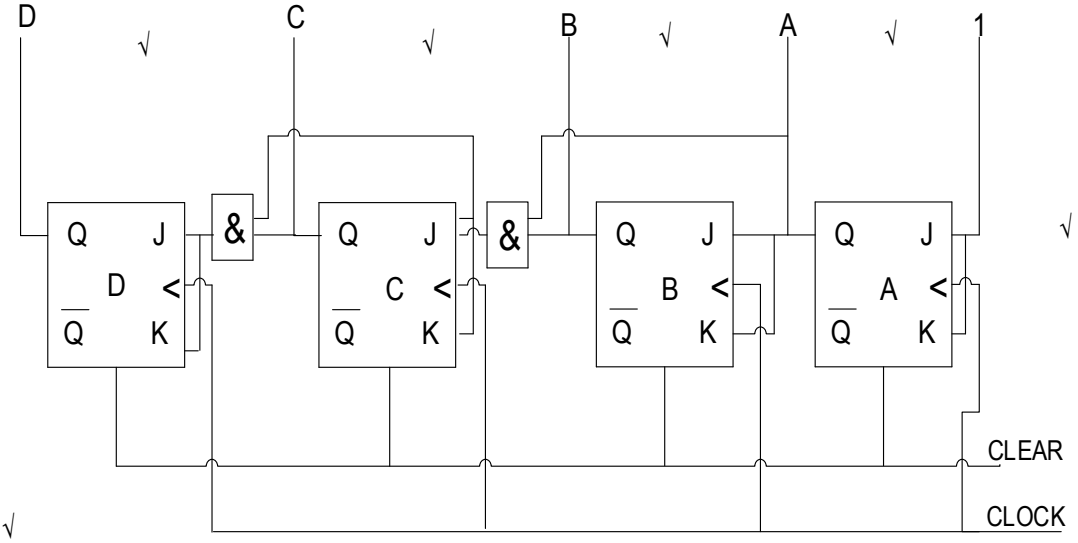
QUESTION 4

4.1



(8)

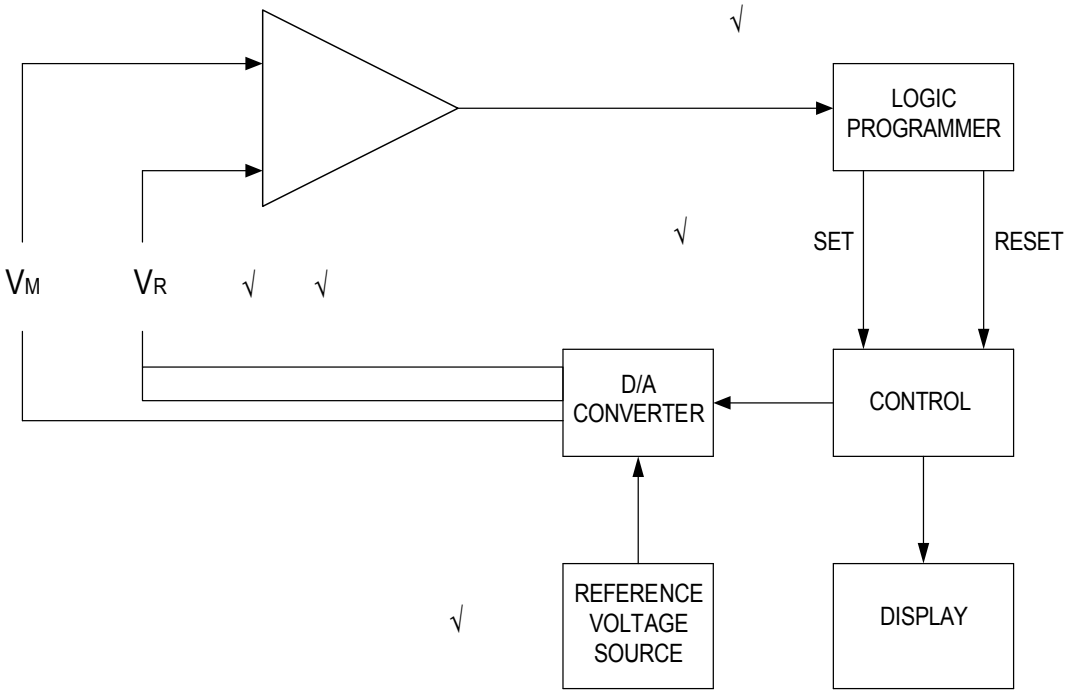
4.2



(6)
[14]

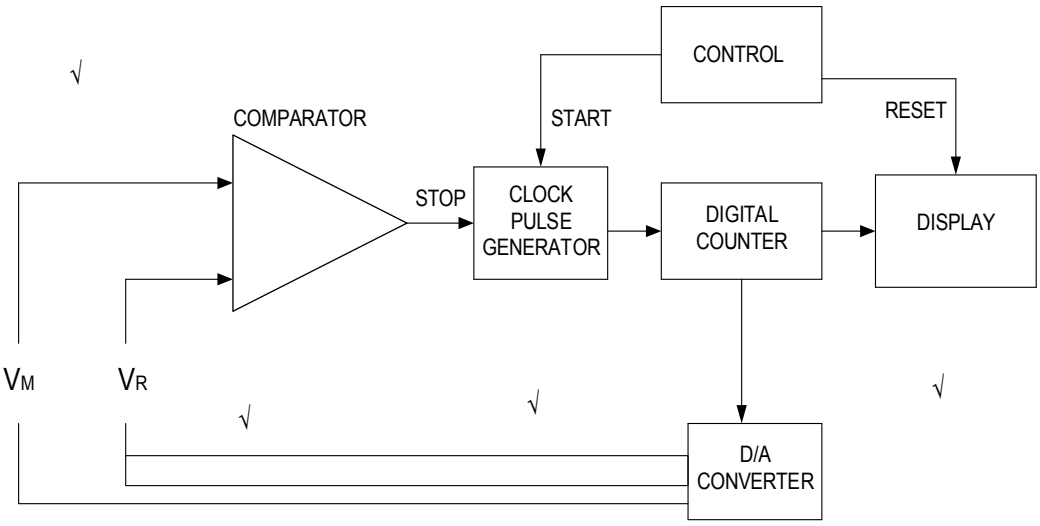
QUESTION 5

5.1



(5)

5.2



V_M = ANALOG VOLTAGE TO BE MEASURED

V_R = REFERENCE VOLTAGE

(5)
[10]

TOTAL: 100

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